



OPEN Neural network approach enhancing churn prediction with categorical encoding and standard scaling

Brijit Bhattacharjee¹, Utpal Madhu⁷, Sutirtha Kumar Guha², Somasree Bhadra³, Pritam Kumar Das⁴, Subham Pankaj Samantaray⁵, M. Zubairuddin⁴ & Shahid Tamboli⁶✉

Customer churn prediction is a crucial application of machine learning in business analytics. This article presents a controlled benchmarking of a multilayer perceptron model trained with one-hot encoding and standard scaling using 10,000 customer records with 12 features. Data pre-processing was performed using one-hot encoding and standard scaling to improve model generalisation. The model achieved an ROC AUC of 0.8640 with a recall of 0.4178 and an F1 score of 0.5534. The measure of correct predictions among cases labelled positive (precision) was relatively high at 0.8214 ± 0.0253 . The moderately low false-positive rate indicates that the model rarely misclassifies non-churners as churners, which is very important for cost-efficient customer retention programs. A group-level heterogeneity review showed that model performance, measured by BRIER scores, was best for France (0.076439), followed by Spain (0.077394), and worst for Germany (0.106267). The model successfully identified the 7 most important fields of judgement, with permutation scores ranging from 0.000147 to 0.120842. The calibration test helps identify underconfident and overconfident model performance through quantile bins. The performance is evaluated using a bin range from 0.12 to 0.25, with a trustworthy prediction for low- to mid-risk customers. A comparative analysis with basis models of machine learning has shown that the best overall accuracy (0.8720) and balanced F1-score (0.6098) were obtained using the Gradient Boosting algorithm, with the best performance, while the proposed Neural Network with categorical encoding and standard scaling attained the highest precision (0.8528), effectively minimising false positives in churn detection. Although the PR_AUC was slightly lower (0.7140), the model had better recall (0.4128) and an equal ROC_AUC (0.8657), indicating the proposed model's strength and ability to operate on complex, high-dimensional churn data.

Keywords Categorical encoding, Churn prediction, Customer behaviour analysis, Data analysis, Neural network, Machine learning, Standard scaling

The loss of a customer or the termination of a customer's relationship with the business is known as customer churn, which is a key issue for business organisations in highly competitive, saturated markets. The telecommunications, banking, and subscription-based e-commerce businesses suffer attrition rates that not only threaten their profitability but also their long-term sustainability. It also has a remarkable financial cost: although winning over a new customer is five to ten times as costly as maintaining an existing one, losing loyal customers erodes brand equity, market share, and revenue stability¹. Telecom churn is around 26–30% annually due to stiff competition, commoditized services, and a series of promotional campaigns launched by competitors^{2,3}.

¹Department of Computer Science and Engineering, Swami Vivekananda Institute of Science & Technology, Kolkata, West Bengal 700145, India. ²Department of Computer Science and Engineering, Meghnad Saha Institute of Technology, Kolkata, West Bengal 700150, India. ³Department of Information Technology, Netaji Subhash Engineering College, Kolkata, West Bengal 700152, India. ⁴Department of Mechanical Engineering, Aditya University, Surampalem, Kakinada 533437, India. ⁵Department of Computer Science and Engineering, Nigam Institute of Engineering and Technology, Govindpur, Cuttack 754006, India. ⁶Department of Mechanical Engineering, Symbiosis International University, Symbiosis Institute of Technology, Pune, India. ⁷Department of Mechanical Engineering, Swami Vivekananda Institute of Science & Technology, Kolkata, West Bengal 700145, India. ✉email: shahidt@sitpune.edu.in

Given such high attrition, churn prediction systems are a priority for telecom operators in customer relationship management (CRM). Companies aim to identify customers who will leave and, through this identification, employ interventions to reduce turnover, potentially through personal offers, service enhancements, or a loyalty program, thus maximizing customer lifetime value⁴. Early research on churn prediction relied on interpretable, simple, and transparent statistical models such as logistic regression, discriminant analysis, and decision trees⁵. Such models, however, were thwarted by the failure to establish the intricate non-linear relationships inherent in customer data. As the quantities of data grew and more variables, such as demographic, behavioural, and transactional ones, the predictive ability of the individual models plateaued⁶. The weaknesses of classical algorithms resulted in the application of machine learning algorithms such as the Random Forest model, Support Vector Machine (SVM), and the application of an ensemble model. Ensemble methods like Random Forests⁷ and Gradient Boosting⁸ capture feature interactions and complex decision boundaries to improve predictive performance. Specifically, XGBoost has become a widely used model for churn prediction because it is efficient and can be regularised. Several studies have shown that it is more effective than baseline classifiers^{9,10}.

Multi-layer Perceptrons (MLPs), Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and hybrid networks, such as BiLSTM-CNNs, have been successfully applied to churn¹¹. The ability of Neural Networks to learn complex dependencies from heterogeneous data makes them more appealing for churn prediction. Nevertheless, churn prediction still has ongoing issues in spite of incredible advances made. The telecom churn data are normally categorical (e.g., type of contract, method of payment, type of internet service, etc.) as well as numerical (e.g., monthly fee, life duration, overall fee). Improper handling of categorical variables - such as arbitrary label assignments - can introduce spurious ordinal relationships, misrepresenting model predictions¹². Similarly, unscaled numerical features can bias training, as attributes with larger magnitudes dominate optimization¹³. The second challenge is class imbalance. Churners usually represent a small minority of customers, typically less than 30% of a dataset. This skew makes the models give high accuracy but low recall on the non-churn class, which is the most common type of data, and that is what the businesses want to recognize¹⁰. Some methods to address this problem include random undersampling, oversampling, and synthetic minority over-sampling technique (SMOTE), all with more or less success¹³. The third issue is associated with interpretability. Although they are very successful in prediction, usually, managers do not understand the process of how it is done, since neural networks are a black box. The methods of interpretability, like feature importance, SHAP (SHapley Additive exPlanations), and LiME (Local Interpretable Model-agnostic Explanations), are becoming commonly used to address this gap¹⁴. Despite extensive prior work on machine learning and deep learning for churn prediction, relatively few studies have systematically examined the impact of pre-processing, especially categorical encoding and feature scaling, on neural network performance. The majority of works focus on the choice of algorithms and do not adequately examine the preparation of diverse data for deep architectures¹⁵. In addition, there is a limited amount of comparative research on preprocessing-enhanced neural networks versus ensemble models such as XGBoost. Table 1 presents the model applied, important findings, and gaps or problems of research. This gives good ground and incentive to declare the way the current research fills these gaps using the advanced neural network methodology and benchmarking.

This research intends to fill the gaps in churn prediction to create a neural network model that has been optimized using categorical encoding and standard scaling to increase the generalization and stability of the model. The proposed model will be benchmarked against the traditional machine learning classifiers, such as Logistic Regression, Random Forest, and XGBoost, to compare the performance and strength of the models. A set of evaluation metrics, comprising accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices, is used in evaluating model outputs to ensure that there is a balanced interpretation of predictive reliability across customer classes. Moreover, the analysis of the importance of features is used to promote interpretability and yield actionable insights to customer relationship management (CRM) applications. Combining the concept of deep learning with preorganized preprocessing and interpretability systems, the study enhances the methodological accuracy of the churn prediction and can provide effective recommendations to organizations that must reduce the effects of customer loss by relying on data-driven decisions taken.

Literature review

Churn prediction is the use of historical customer data, which may be both demographic and behavioral, and also transactional, to predict future customer trends resulting in customer exit²⁸. The list of ways to do so is extensive, and the classic statistical solutions reach the state of sophisticated Machine Learning (ML) and Deep Learning (DL) approaches²⁰. The appropriate methodology will be selected will generally be selected based on the characteristics of data, such as its dimensions and variability, and organizational uses, such as interpretability, computational efficiency, and predictability¹¹.

Early statistical approaches

At the beginning of research on churn prediction, most studies relied on a statistical method. Logistic regression was used to give a probabilistic model of the dichotomous churn and non-churn customer, which is most valued for its being easy to interpret and understand¹⁶. Rule-based segmentation was provided by decision trees, and this makes forecasts easy to understand by managers and practitioners. This methodology was improved through survival analysis that provides time-to-event data modeling, which predicts the likelihood of churn in specified intervals². Although such performance techniques provided background information, they frequently assumed a linear relationship and independence of the variables, which limits their ability to reproduce customer responses in a complex way⁵. This resulted in a bad prediction of large and varied data.

Research Study & Year	Model Used	Findings by Researchers	Research Gaps / Related Issues
Ngai et al., 2009 ¹⁶	Logistic Regression, Decision Trees	Interpretable, transparent but weak with complex non-linear data	Poor at modeling complex behaviors, limited scalability
Jahromi et al., 2010 ²	Survival Analysis	Modeled time-to-churn, helpful in telecoms	Still assumes feature independence, limited for high-dimensional data
Breiman, 2001 ⁷	Random Forest	Reduced variance, handled non-linearity, and better accuracy	Limited interpretability overlooks the preprocessing of data
Friedman, 2001 ⁸	Gradient Boosting	Incremental learning, strong predictive performance	Complex, can be sensitive to outliers and class imbalance
Chen & Guestrin, 2016 ⁹	XGBoost	High performance, regularization, industry standard	Data preparation and feature scaling underexplored
Goodfellow et al., 2016 ¹⁷	Deep Learning (NN)	Captures hierarchical, complex patterns, robust for big data	Black-box, interpretability, training complexity
Banu et al., 2022 ⁴	AI-Based ML Models	Improved retention through AI-driven insights	No detailed feature scaling/coding, deployment issues
Alghamdi & Al-Baity, 2022 ¹²	Augmented Analytics, DL	Enhanced accuracy using advanced analytics and encodings	Little attention to high-cardinality encoding/scaling
Liang, 2025 ¹⁸	One-hot Encoding for Categorical Variables	Demonstrated that one-hot encoding effectively represents nominal categorical features without imposing artificial ordinal relationships, enabling ML models to process non-numeric churn attributes	High-cardinality features cause dimensionality explosion; scalability and integration with neural network embeddings remain limited
Mirkovic et al., 2022 ¹⁹	Invoice Data ML	Modeled B2B churn, extended scope beyond B2C telecom	Non-contractual data harder to model, limited feature prep
Sahar, 2018 ²⁰	Various ML Models	Systematic comparative study improved recall/accuracy	No focus on preprocessing, scaling, open-source telecoms focus
Malikireddy & Kasa, 2021 ²¹	Various ML Models	Systematic review highlighted best performing models	Opaque findings, few dealt with class imbalance
Ebrah & Elnasir, 2019 ⁶	ML, Recommendation	ML boosted prediction accuracy with customized plans	Marginal attention to precise encoding/scaling
Liu & Zhuang, 2015 ¹	Cost-sensitive Segmentation	Churn prediction linked to misclassification costs	Less on advanced DL/ensemble benchmarking
Vafeiadis et al., 2015 ⁵	Multiple ML Models	Compared performance, highlighted tradeoffs in interpretability vs. accuracy	Insufficient preprocessing, mostly legacy ML models
Dhotre et al., 2023 ³	Customer Retention Models	Emphasized targeted interventions for telecom sector	Deployment and business constraint discussions lacking
Bhuria et al., 2025 ²²	Ensemble Voting Classifier	Improved bank client retention with behavioral data mix	Handling of non-numeric features with mixed-value aggregation
Sikri et al., 2024 ¹⁰	Hybrid ML Models	Hybrid approaches achieved high accuracy	Did not rigorously address imbalanced data, encoding
Khattak et al., 2023 ¹⁰	BiLSTM-CNN, Deep Learning	Surpassed traditional ML, effective for sequential behavior modeling	Lacked transparency/interpretability
Srinivasan et al., 2023 ²³	RF + SMOTE-ENN, Various ML	Highest F1-score (95%) via balancing, highlights ensemble-sampling value	Focused on score, less on explainability and scaling
Chinnaraj, 2023 ²⁴	Bio-inspired Optimization, NN	Metaheuristics with DL improved optimization/ performance	Lacked real-time, deployment references
Matteucci et al., 2023 ²⁵	Encoder Benchmarks	Benchmarked categorical encoders, proved encoding method changes performance	Few studies combine encoding with scaling
Peng et al., 2023 ¹⁴	XAI, SHAP and LIME	Provided interpretability to black-box churn models	Limited to post-hoc analysis, not integrated to preprocessing
Imani et al., 2025 ²⁶	ML/DL Comparative Review	Surveyed advances/trends in churn modeling	Gaps in business deployment and feature preprocessing
Ong et al., 2024 ¹³	Resampling, Scaling	Highlighted resampling and scaling for fairness/optimization	Rarely joint encoded/scaled neural networks
Chang et al., 2024 ²⁷	ML for Telecoms	Compared ML models for telecom churn, validated ROC and F1 differences	No systematic preprocessing exploration
Li et al., 2024 ²⁸	Data Science/AI	Cited value of ML/AI for marketing retention	Few controlled benchmarks against DL
Chajia & Nfaoui, 2024 ¹⁵	LLM Embeddings, Logistic Regression	Used modern representations (embeddings) for improved logistic regression performance	No hybrid NN/ensemble benchmarking
Maan & Maan, 2023 ²⁹	Explainable ML	Used explainability for churn prediction	Focused on XAI, less on preprocessing methods
Alotaibi & Haq, 2024 ³⁰	ML/Ensemble in Telecom	Ensemble models boosted telecom churn accuracy	Did not jointly optimize preprocessing for NN
Hossain & Islam, 2023 ³¹	Hybrid ML, Feature Selection	Feature engineering and preprocessing significantly improve neural network learning and pattern extraction	Limited focus on churn-specific datasets
Huynh et al., 2022 ³²	Artificial Neural Networks	Feature scaling ensures balanced learning, stable convergence, and improved prediction accuracy	Application not validated on categorical-heavy telecom data
Liu et al., 2024 ³³	Hybrid Neural Networks	Adaptive neural networks improve churn prediction accuracy under dynamic data streams	High computational complexity, limited explainability
Srikanth et al. 2026 ³⁴	Convolutional neural networks	It proposed an IoT intrusion detection system using a hybrid WOA-PSO algorithm for feature selection combined with a CNN	Its adaptability to zero-day attacks, robustness against adversarial threats, and effectiveness in real-world IoT environments require further investigation
Poojitha et al. 2025 ³⁵	Support Vector Classifier	It predicted customer churn in subscription-based telecom services using machine learning techniques.	Issues such as class imbalance handling, real-time churn prediction, feature interpretability, and real-world datasets are not addressed
Santos et al., 2023 ³⁶	ML on Tabular Data	Independent validation datasets improve robustness and generalization assessment	Not specific to telecom churn behavior
Continued			

Research Study & Year	Model Used	Findings by Researchers	Research Gaps / Related Issues
Vaiyapuri et al., 2024 ³⁷	DL + Feature Engineering	Feature transformation reveals hidden patterns and enhances predictive robustness	Focused on security datasets, not customer churn
Yan & Resnick, 2023 ³⁸	Bayesian Encoding Models	Target-based encoders handle high-cardinality categorical features efficiently	Limited evaluation with deep neural networks

Table 1. Summary of key research studies on customer churn prediction: models used, findings, and gap identification.

Emergence of machine learning and ensemble methods

Collaborative methods were introduced to overcome the limitations of single classifiers. Breiman⁷ has shown that forecasting was greatly enhanced by the use of Random Forests through bagging (bootstrap aggregation) of decision trees, significantly improving predictive performance by reducing variance. On the same note, Friedman⁸ proposed the idea of Gradient Boosting, which incrementally builds strong learners by correcting the residual errors of prior models. Gradient Boosting was also suggested to be used on churn prediction tasks, where XGBoost⁹ outperforms other Gradient Boosting models by being scalable and regularized. Secondary research demonstrates that XGBoost is more effective than individual classifiers, in particular, it is more effective in imbalanced datasets¹⁰. Hybridization models that combine multiple algorithms have been shown to perform highly in predictive tasks, which include: SvmAda, RfAda, KnnSgd, and others. As an example, KnnSgd, a hybrid of K-Nearest Neighbors and Stochastic Gradient Descent, achieved up to 92.42% accuracy, demonstrating the merit of such a mix of the complementary learning policies¹⁰. Ensemble ML models have been especially beneficial in banking and telecommunication sectors, where they significantly help make churn predictions more accurate and apply them in retention interventions²².

Deep learning in churn prediction

Neural networks offer superior capacity for non-linear modeling of high-dimensional interactions than traditional ML. Goodfellow et al.¹⁷ highlighted the strength of deep architectures in learning hierarchical representations. Combination with metaheuristic optimization, including Elephant Herding Optimization, has increased the neural network performance by optimizing network parameters and choosing the best features to enable less computational complexity and higher accuracy of predictions. Proactive churn management is possible through these advances, where the organization is able to generate an intervention before the customer becomes disengaged³⁹. In practice, Multi-Layer Perceptrons (MLPs) were extensively applied to churn prediction, which has been averagely successful in processing demographic and transactional data²¹. Further sophisticated architectures, such as RNNs and BiLSTM-CNNs extend this capability by capturing sequential dependencies in customer behavior¹¹. To take an instance, a BiLSTM layer can resolve the issue of order in service utilization (temporal), whereas a CNN can accomplish local patterns (analysis), together enhancing predictive accuracy. These benefits are proven through empirical studies. Khattak et al.¹¹ reported that the BiLSTM-CNN had a significant marginal out-performance to both logistic regression and decision trees when used on telecom data. Likewise, Srinivasan et al.²³ indicated that the Random Forest model with the use of SMOTE-ENN showed the best performance of 95% of the F1-score.

Explainable AI and augmented analytics

With the increasing complexity of predictive models, for managers to make decisions on them, interpretability is more important. SHAP and LIME are some of the explainable AI (XAI) techniques that help understand why a customer is churning and what factors contribute the most to the prediction decision^{27,29}. Augmented analytics platforms further support churn prediction by combining automated insights, visualization recommendations, and “what-if” scenario simulations. These tools enable rapid model deployment and refinement, improving operational efficiency and accessibility for non-technical stakeholders¹².

Categorical encoding techniques

Encoding of categorical features is one of the foundations of churn prediction. The majority of Machine Learning models are unable to deal directly with non-numeric variables. Nominal variables are usually represented using one-hot encoding, which creates binary markers of each category, which avoids contrived ordinal assumptions^{6,18}. Nevertheless, it makes features such as location or methods of payment very high-cardinality, which is a significant growth in dimensionality. In comparison, label encoding gives categories integer values, and is applicable in ordinal variables like terms of a contract. However, when used inappropriately on nominal data, it could lead to the creation of spurious ordinal correlations^{10,26}. New ways, like target encoding and embeddings, have been introduced as an alternative. Target encoding instead of categories uses average churn probabilities, which are more compact representations; embeddings, commonly found in deep learning, are a way to map categories into dense vectors in a low-dimensional space, allowing a model to learn similarities between semantically related things¹⁵. Bayesian-inspired ones (e.g., M-Estimate, James-Stein) generate smaller yet informative representations, which reduces the curse of dimensionality and improves the efficiency of the model^{25,40}. These methods have a great influence on the model¹², pointed out that categorical features that have been poorly encoded may cause multicollinearity or distortions in the feature distributions, which reduces the accuracy. Therefore, it is essential to select a suitable encoding technique to have a powerful churn forecast.

Feature scaling and normalization

Numerical feature scaling makes sure that the variables with different magnitudes make an equal contribution to the model optimization. In the absence of scaling, the features of total charges (which often have thousands of values) can easily dominate the features of tenure (measured in months), which will bias the optimization process. The most common technique is standard scaling, which normalizes the variables to mean 0 and variance 1³⁰. Other approaches, such as min-max scaling, where the values are normalized to [0,1], and robust scaling, where the outliers are minimized, are also possible. Z-score normalization (also known as standard scaling) and other methods, including min-max scaling and robust scaling, enhance the convergence and predictive power and accuracy of the neural networks used on churn data¹³.

Applications across industries

Even though telecom has been the main area of concern, churn prediction applies to various industries. In the banking industry, ML models have also improved retention of clients since they predict disengagement more effectively than traditional ones²². Predictive models are used in the context of e-commerce to make personalized recommendations and loyalty campaigns. Churn models are also used by the energy companies to reduce customer switching in the deregulated market⁴¹. This cross-industry usage points to the applicability of churn prediction structures, but each industry has its own challenges. As an example, in B2B, it is a situation where the analysis of invoice-level data is required, which is more complex¹⁹.

Research gaps

There are still a number of gaps, even though progress has been made. To begin with, ensemble and deep learning approaches have high predictive accuracy; however, the impact of preprocessing, especially encoding and scaling, has not been explored. Second, the majority of research pays attention to accuracy and ignores recall, which is an important parameter in detecting churners. Third, there is also an issue of interpretability, where managers need to have actionable predictions instead of raw predictions. The existing research fills these gaps with a systematically integrated categorical encoding and standard scaling into a neural network structure, performance comparison of classical and ensemble algorithms, and interpretation analysis. Such a strategy is a guarantee of methodological innovativeness and practical applicability of CRM strategies.

Methodology

The proposed work aims at transforming raw customer data into useful features that could be used to feed the model before training and prediction. Each of the steps is discussed in greater detail below, starting with feature selection and ending with hyperparameter tuning, and Fig. 1 provides a visual representation of the whole process.

Dataset

The Kaggle dataset (<https://www.kaggle.com/datasets/gauravtopre/bank-customer-churn-dataset/data>) is a collection of approximately 10,000 records, which has 12 rows. These are 12 features columnized. Of all the features, the credit score is critical in the quantification of bank customers. It lies between the range of 350 and

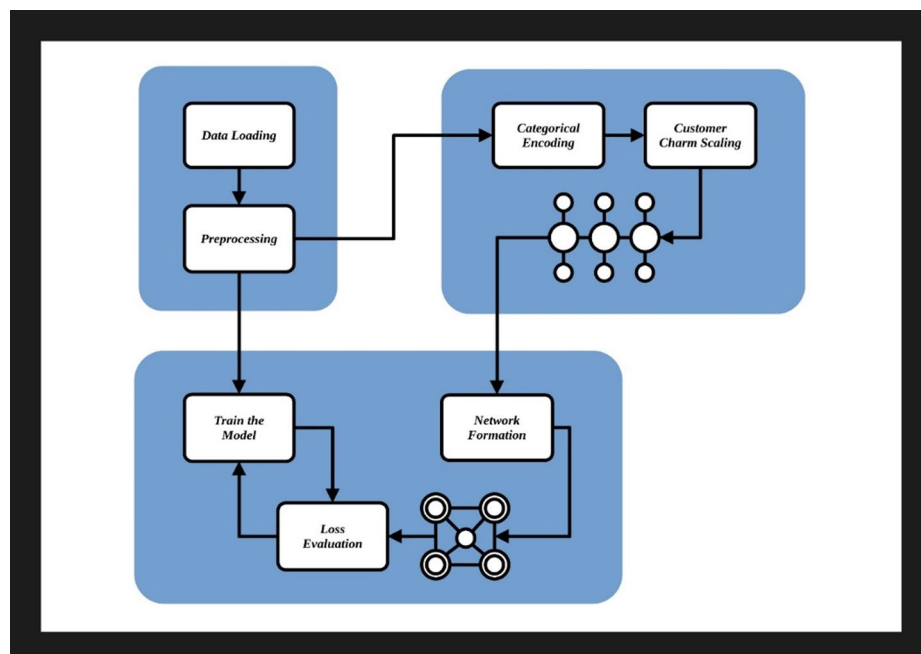


Fig. 1. The system framework.

850. The country attribute is shown to have records of three countries, namely 5,014 records of France, 2,509 of Germany, and 2,477 of Spain. The column of balance contains the bank account balance that may reach 25 lakhs. The data on bank accounts are shown in the range of 18–92. Churn prediction also relies a lot on the tenure column, which will explain the time taken by the customer with the bank. The tenure aids in the recognition of new and old customers. It can be used in customer survival plots, active-user rates, and cohort benchmarking. The other significant aspect is the estimated salary. Reduced pay also brings on the affordability pressure, and this raises the churn. Extremely remunerated staff are also associated with an increase in work turnover.

Data preprocessing

The dataset is publicly accessible and commonly used in churn prediction studies. After data collection, there is a series of steps involved in data preprocessing. This step implements missing value imputation to categorical encoding. Normalization is carried out to encode the data in the neural network efficiently. The final step of this process is referred to as data cleaning. Outliers and inconsistencies are detected and handled to ensure data quality. One-hot encoding is used to transform categorical variables. The banking dataset maintains large data distributed among three categories: France, Germany, and Spain in the country attribute. One-hot encoding is an appropriate design choice for such a high-cardinality categorical attribute. While one-hot encoding guarantees methodological reproducibility through a baseline, target encoding uses a stringent training fold of a cross-validated pipeline for target leakage. The adopted encoding mechanism simplifies data spaces, condensing a multidimensional format into a unidimensional one. The process also enhances compatibility across various modelling techniques, emphasizing specific categories under unique labels. In the banking dataset, one class has more samples than another, like the negative churn have higher occurrence chance than the positive class. In pre-processing, ignoring the minority class forced the model to be biased towards specific classes, leading to class imbalance. The proposed work was implemented in cross verification framework that applies stratified splits on all training, validation, and test partitions to ensure original class proportions. Imbalance-aware loss reweighting is applied to linear models during training, where misclassification obtains a higher penalty than non-churners. Gradient boosted models include built-in class weighting as a tuned parameter. In a neural network, weighted cross entropy applied to the minority class to deal with imbalance during training and optimization.

Feature engineering

Feature engineering is decisive for developing the model's capability to obtain complex customer behaviors. The proposed approach includes interaction features and dimensionality reduction. In interaction features, new variables are generated by taking the product of variables (e.g., tenure and monthly fees) to capture non-linear interactions.

Baseline for the proposed model

The baseline of the proposed model was designed using 300 estimators in the form of a tree, a learning rate of 0.7 while training XGBoost. Sub samples are spread with 0.9 sub and column samples. The objective function maintains binary logistics while maintaining tree methods. Other models, like light-GBM performs training with 0.05 learning rate and 400 tree estimators. Optimization carried out through a randomized search.

Detailed workflow of the proposed churn prediction model

The Activation-Driven Multilayer Neural Network algorithm is based on the concept that processed encoded features are passed through a series of hidden layers sequentially to maximize the accuracy with which churn is classified, as outlined in Algorithm 1. Raw input data is accumulated in the data collection. Collected data is cleaned and standardized in a pre-processing state. New features are created, and dimensions are reduced in feature engineering. Data is divided into training, validation tests, etc., in the data splitting part. The sequential process of our proposed methodology is depicted in Fig. 1. Before feeding the input feature into the neural network, each neuron computes a weighted sum of the inputs from the previous layer. Learned parameters (N'), with input value (I) and a bias S' performs forward propagation (D') in Eqs. (1),

$$D' = N' I + S' \quad (1)$$

Forward propagation is carried out with 30 input features $I \in \mathbb{R}^{30 \times 1}$ in Eq. (1), upon the first dense layer. The output from each layer is a linear function of the input, which limits the network to learn complex patterns. To predict churn where multiple factors interact in a complex way, each layer must transform data in a meaningful way. This enables such multi-layer networks to adopt activation functions. The activation operation is illustrated via Eqs. (2),

$$R' = P' (D') = \max (O, D') \quad (2)$$

In Eq. (2), ReLU activation is represented by the variable R' . At layer P' the result from the forward propagation allows the network to learn intricate relationships between features. A formative architecture of the entire process is depicted in Fig. 2. At layer 1, each neuron evaluates output as,

$$d'_j = \sum_{i=1}^v n'_{ji} q_i + s'_j \quad (3)$$

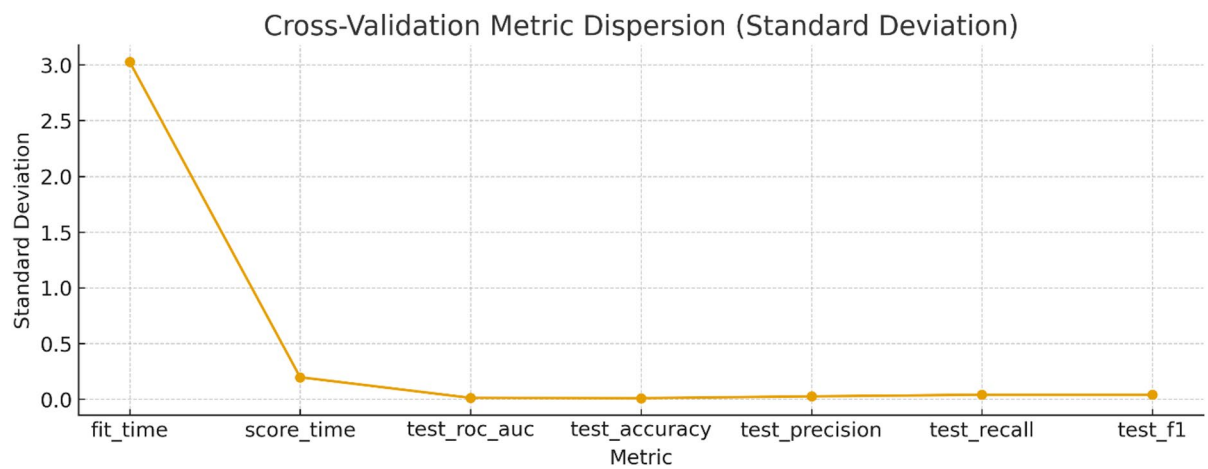


Fig. 2. Cross-validation performance metrics showing mean and standard deviation across K-folds.

In Eq. (3), input features (q_i) are multiplied by weights (n'_{ji}) to produce a summation up to v steps with j starting from 1 to 32. Like Fig. 1, the churn scaling data from Eq. (3) passed through the activation function via Eqs. (4),

$$r'_j = R(d'_j) = \max(O, d'_j) \quad (5)$$

```

1  Input: Dataset  $I$  where  $I \in \mathbb{R}^{\wedge} (30 \times 1)$  a
    30-dim feature vector
2  Output: Churn probability function  $\hat{c}$ 
4  .....
    .....
5  Begin
6  | Initialize the  $D' = N' I + S'$ 
7  | If  $R' = \max(O, D')$ 
8  |   Repeat
9  |     
$$d'_j = \sum_{i=1}^v n'_{ji} q_i + s'_j$$

10 |   End
11 | For each neuron, select  $d'_j$ 
12 |   Formulate,
    |     
$$d^2_j = \sum_{j=1}^{32} n_{Kj}^2 9'_j + s_K^2$$

13 |   End
14 | End
15 | For  $r_K$ 
16 |   If  $r_K^2 = \max(O, d_K^2)$ 
17 |     Repeat
    |       
$$L(N, S) = -\frac{1}{m} \sum_{i=1}^m [c^i \log(\hat{c}^i + (1$$

    |         
$$- c^i) \log(1 - \hat{c}^i))]$$

18 |     End
19 |   End
20 |   Final solution  $\hat{c} = P^3(d^3)$ 
21 End

```

Algorithm 1. Activation-Driven Multilayer Neural Network algorithm for customer churn estimation.

At layer 2, all the data, along with numerical values transferred from hidden layer 1, passed through Eqs. (5) and (6), numerical steps.

$$d^2_j = \sum_{j=1}^{32} n_{Kj}^2 9'_j + s_K^2 \quad \forall K = 1, 2, \dots, 16 \quad (5)$$

$$r_K^2 = R(d_j^2) = \max(O, d_K^2) \quad \forall K = 1, 2, \dots, 16 \quad (6)$$

From hidden layer 2 to the output layer, the computation is carried out as,

$$d^3 = \sum_{K=1}^{16} n_K^3 q_K^2 + s^3 \quad (7)$$

During training, a numerical evaluation is carried out during network formation, along with loss evaluation, as represented in Fig. 1. The framework depicts the sequential stages of prediction, including data preprocessing, categorical encoding, and churn scaling. The fine-tuned neural system systematically improves the objective function to enhance outcome reliability with respect to customer retention outcomes. At this step, the objective of the loss function can only be achieved via probabilistic customer churn ($\hat{c}$) evaluated through the sigmoid function as,

$$\hat{c} = P^3(d^3) = \frac{1}{1 + e^{-d^3}} \quad (8)$$

Loss function $L(N, S)$ during training, evaluated via Eq. (9) for all training samples,

$$L(N, S) = -\frac{1}{m} \sum_{i=1}^m [c^i \log(\hat{c}^i) + (1 - c^i) \log(1 - \hat{c}^i)] \quad (9)$$

In Eq. 9, c^i represents the true churn label with $\hat{c}^i$ for predicted churn and m number of training examples in S .

The framework in Fig. 1 starts with gathering and importing data, where raw customer data is cleansed to ensure quality and consistency. Feature encoding transforms categorical inputs into usable numerical values, and cross features are engineered to model multidimensional dependencies within the data. The data is provided to a multi-tier neural network model. During optimization, the model successively updates its parameters to minimize error and increase the precision of predictions.

Hyperparameter tuning

The churn prediction was carried out through nested cross-validation, where the 10,000-record Bank customer dataset was split into training and test sets multiple times, recorded as iterations. During each iteration, the model records performance for each segment until the entire dataset has been covered. The data partitioning was carried out using an outer stratified k-fold procedure with 3 iterations. Each outer split preserves churn or non-churn class distribution applied to tune hyperparameters at the second level of cross-validation. The Randomised search operation predefines the search space as an inner stratified K-fold within each outer training split. It samples distributions with a size of 25 for each model per outer fold. The mean ROC-AUC evaluate sampled configuration across the inner folds based on the highest ROC-AUC chosen from the outer training set. The score of the outer fold for each model can be averaged to obtain the overall performance, which acts as an input to the Friedman test to compute calibration metrics and bootstrap confidence intervals. The pre-processing was embedded inside the cross-validation pipeline, where numerical features were processed with median imputation and categorical features by one-hot encoding. The pre-processing prevents information leakage by scaling statistics into fold-consistent transformations. The tuning design applies a subsampled class-weighted scheme to train using Random Forest. XGBoost search space maintains DART-style dropout configuration using booster, rate-drop, and skip-drop parameters.

Results and discussion

Cross-validation performance

The model achieved a mean and ROC_AUC of 0.8640 ± 0.0119 , accuracy of 0.8630 ± 0.0092 , precision of 0.8214 ± 0.0253 , recall of 0.4178 ± 0.0397 , and F1-score of 0.5534 ± 0.0397 . The cross-validation chart helps identify the configuration with the best generalisation performance. The entire performance was recorded through a chart in Fig. 2. Such findings mean that the model has good general classification performance and strength, as shown by the large accuracy and ROC_AUC. Nevertheless, the relatively lower recall indicates that although the model is successful in detecting a majority of the non-churn, it can still fail to detect some real churn.

Model comparison

A comparative analysis of several customer churn prediction machine learning models is presented in Table 2, with emphasis on ranking performance (ROC_AUC, PR_AUC) and classification performance (Accuracy, Precision, Recall, and F1-score). The Gradient Boosting algorithm achieved the highest accuracy (0.8720) and a better F1-score (0.6098), indicating a balanced approach to precision and recall. With the highest accuracy of 0.8660 and precision of 0.8528 - the highest precision of all models - categorical encoding, along with linear choice of a standard scaling, resulted in the Proposed Neural Network Model of 0.8528 precision. This implies that the model has a high ability to reduce false positives, which is vital in retaining good customers during churn prediction activities. Yet, though the proposed model has a slightly lower PR_AUC (0.7140) and a similar ROC_AUC (0.8657), making it stronger in ranking and discriminative, the comparison between the proposed model and the Gradient Boosting model reveals that the former has a higher recall (0.4128) than the latter. The Random Forest and XG-Boost were also competitive options, achieving accuracies of about 0.8670 and

Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
Random Forest	0.8670	0.8219	0.4423	0.5751	0.8603	0.7010
XGBoost	0.8665	0.7713	0.4889	0.5985	0.8571	0.7003
LightGBM	0.8590	0.7341	0.4816	0.5816	0.8535	0.6917
Extra Trees	0.8560	0.7598	0.4275	0.5472	0.8453	0.6544
K-Nearest Neighbors	0.8425	0.7738	0.3194	0.4522	0.8172	0.5924
Gaussian NB	0.8060	0.5331	0.3759	0.4409	0.7773	0.5204
Logistic Regression	0.8095	0.6250	0.1597	0.2544	0.7751	0.4770
Decision Tree	0.7945	0.4954	0.5307	0.5125	0.6963	0.3584
Gradient Boosting	0.8720	0.8032	0.4914	0.6098	0.8712	0.7220
Proposed Model	0.8660	0.8528	0.4128	0.5563	0.8657	0.7140

Table 2. Evaluating ranking quality and operating characteristics across models.

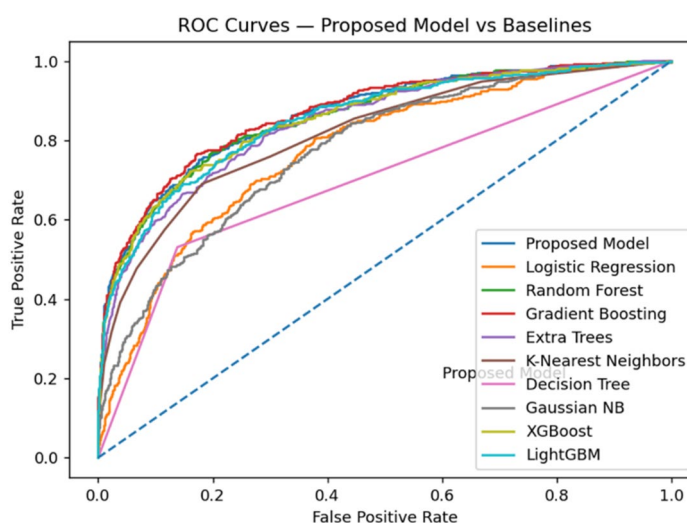


Fig. 3. ROC curves comparing the ranking performance of all models.

0.8665, respectively, but with lower balanced F1-scores. On the other hand, the Classic models, such as Logistic Regression and Gaussian NB, had low predictive ability due to their linear assumptions.

In Fig. 3, the performance differences across models, shown as ROC curves, are relatively small and vary with the decision threshold. Since churn data are imbalanced, it is important to consider PR-AUC, which is reported in Table 2. The proposed model achieves one of the highest PR-AUC values, demonstrating efficient identification of churners relative to the number of alerts generated. In Fig. 4, high precision at low recall for the positive (churn) class, while the negative class shows a broader curve, indicates that the model ranks true churners well, but loses precision as it predicts more positives. In addition, a baseline precision line indicates performance above chance. Despite class imbalance, the figure shows that pre-processing steps, such as categorical encoding and standardisation, improve the model's ability to distinguish churners from non-churners.

Confusion matrix analysis

Figure 5 is the confusion matrix that summarises the performance of the proposed churn prediction model, which displays the true/false distribution of churn and non-churn. The model produced 1564 True Negatives, 168 True Positives (TP), 29 False Positives (FP), and 239 False Negatives (FN). The false-positive rate is relatively low, which indicates that the model is not more likely to classify non-churners as churners, of great importance in cost-effective customer retention programs. False negatives, however, mean that there are cases of actual churners not caught, which can be mitigated by adding more feature representation or increasing the threshold. All in all, the model is highly sensitive and specific and therefore highly predictive with respect to customer segments.

Feature importance

Permutation-based feature importance in Table 3 highlights balance and active member as the key predictors of customer churn. Credit card, country (France), and age also contribute positively, while estimated salary and number have little impact when controlling for other variables. Thus, financial engagement and customer activity are the main predictors of churn.

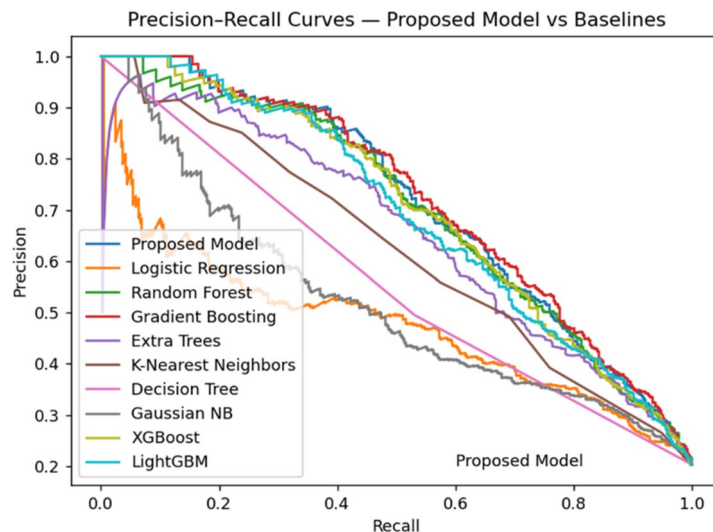


Fig. 4. Precision–Recall curves highlighting class imbalance and ranking efficiency.

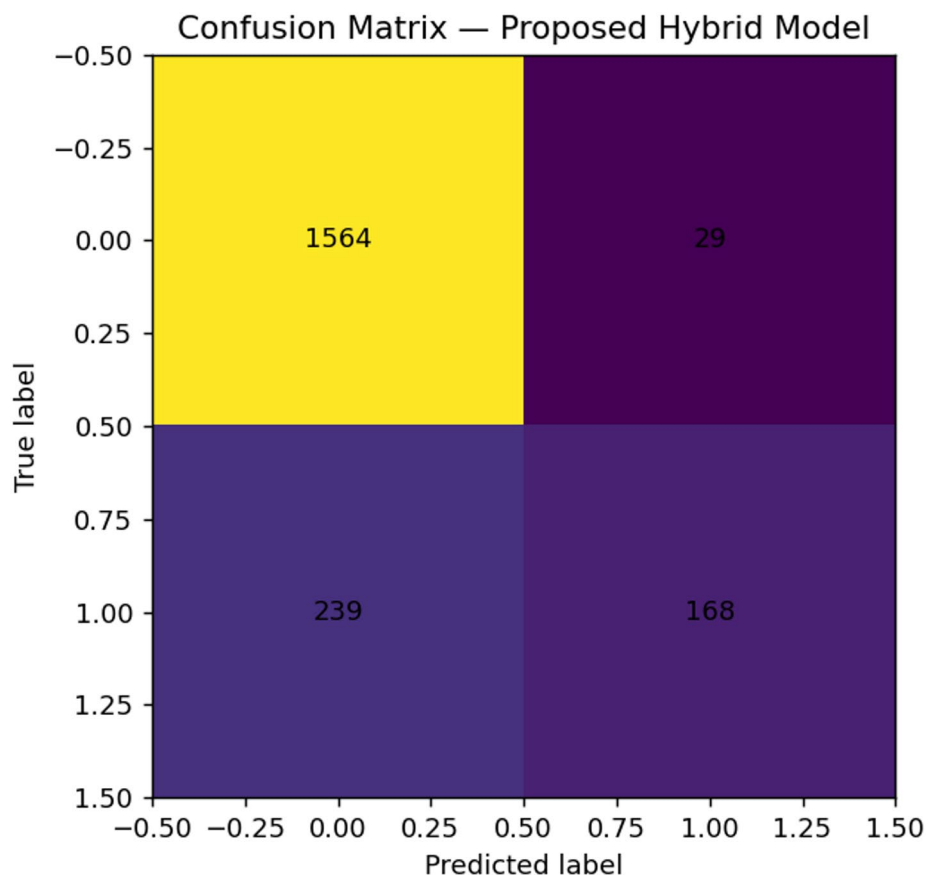


Fig. 5. Confusion Matrix illustrating the proposed model's label prediction accuracy.

Calibration and profit optimisation

Figure 6 shows a calibration assessment of our trained model to evaluate its performance in selecting customers likely to churn. The blue dashed line indicates ideal calibration and helps distinguish between underconfident and overconfident model predictions. If the true rate is less than the predicted outcome, the points below mean indicate over-confident behaviour. The orange curve tracks the outcome in 10 quantile bins. The bins within

Features	Importance (mean)	Importance (standard deviation)	Permutation-based importance
Balance	1.208418×10^{-1}	0.005583	0.120842
Active_member	1.178130×10^{-1}	0.009871	0.117813
Credit_card	3.678625×10^{-2}	0.005381	0.036786
Country_France	3.340120×10^{-2}	0.003834	0.033401
Age	2.316585×10^{-2}	0.002043	0.023166
Tenure	7.892021×10^{-3}	0.002139	0.007892
Credit_score	1.472968×10^{-4}	0.001556	0.000147
Estimated_salary	-4.627123×10^{-7}	0.000344	0.000000
Products_number	1.208418×10^{-1}	0.000846	-0.000645

Table 3. Permutation-based feature importance for the top nine drivers of churn.

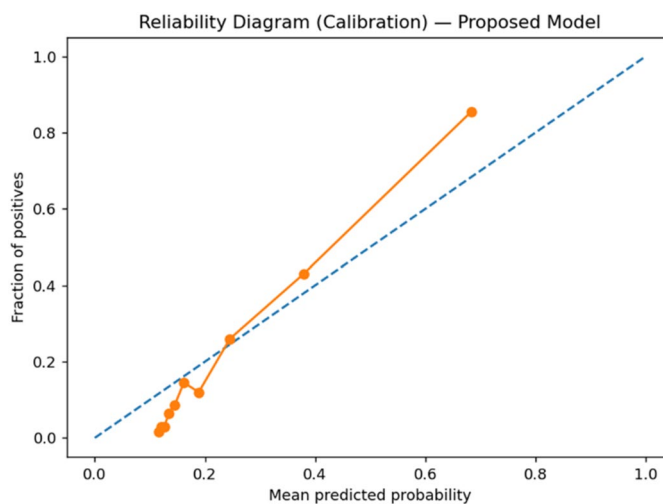


Fig. 6. Calibration performance of the proposed Churn Prediction Model.

the range of 0.12 to 0.25, plotted very close to the ideal calibration line, indicate that the model probabilities are trustworthy in terms of small campaigns, as they produce low to mid risk.

Bins within the range of 0.38 to 0.40, lying above the calibration line, indicate under-confident behaviour, meaning the model predicted a churn rate of 43% while the mean prediction stands between 38% and 40%. The highest bins from 0.67 and above indicate conservative behaviour at the very top end. Equation 10 represents the formula for the *Brier* score, where the predicted probability (P_h) is subtracted from the actual outcomes (A_h) out of H samples.

$$Brier = \frac{1}{H} \sum_{h=1}^H (P_h - A_h)^2 \quad (10)$$

The Expected Calibration Error (ECE) measures the model's performance by comparing observed outcome frequencies across the dataset, providing a single summary metric for calibration quality.

$$ECE = \sum_{g=1}^G \frac{s_g}{H} |f_g - t_g| \quad (11)$$

Equation 11, compute ECE for each bin (g) with binsize (s_g). Fraction of positives (f_g) subtracted from the mean predicted probability (t_g) for total G bins. Figure 7 illustrates an experiment designed to sequence customers from the highest to the lowest churn possibility. The targeted fraction in the x-axis represents customers that the model wants to target with the highest risk. The vertical axis represents the profit the organisation can gain through those customers via the retention offer. From 0.1 to 0.3, the model identified customers with the highest profit potential, and the curve peaks at a high level. After that range, the curve started to drop. After 0.3, the curve declines due to an increase in campaign cost, indicating that the organisation must avoid those customers because they lead to a loss of money.

Group-level heterogeneity

In Table 4, ROC_AUC 0.94 demonstrates the discriminative capabilities of the proposed model across countries and gender. The highest ROC_AUC of 0.953355 and precision 0.968421 of Spain indicate a strong ability to

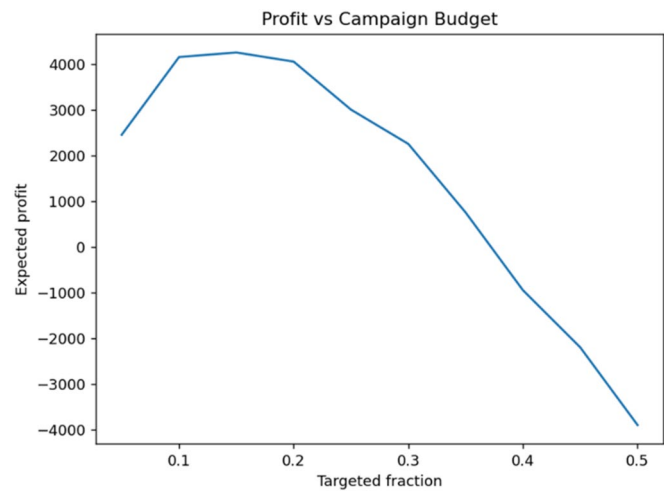


Fig. 7. Operational Decision Curve to identify cumulative expected profit across increasing targeting fractions of the economically optimal customer list.

Segment	Level	Number of records	ROC_AUC	PR_AUC	Precision @0.5	Recall @0.5	Flag @0.5	Brier
Country	France	5014	0.945585	0.831034	0.946237	0.434568	0.595601	0.076439
Country	Germany	2509	0.951318	0.912104	0.913043	0.619165	0.737921	0.106267
Country	Spain	2477	0.953355	0.855946	0.968421	0.445521	0.610282	0.077394
Gender	Female	4543	0.944946	0.882140	0.932722	0.535558	0.680424	0.095941
Gender	Male	5457	0.954001	0.851808	0.934783	0.478842	0.633284	0.074351

Table 4. Segment-wise heterogeneity analysis across country and gender.

accurately classify churn and non-churn customers, with minimal false positives. Germany maintains a superior overall ranking and probabilistic calibration, with PR_AUC of 0.912104 and ROC_AUC of 0.951318. France obtained 0.945585 ROC_AUC and 0.946237 Precision. Germany outperforms all other countries, scoring 0.619165 and showing greater sensitivity in identifying actual churners. In contrast, Spain (0.445521) and France (0.434568) exhibit moderate recall. Moreover, Germany is the highest FLAG of all (0.737921) and marginally greater Brier (0.106267), implying a trade-off between good recall and slightly lower calibration accuracy. The Male group has a better ROC-AUC (0.954001) and precision (0.934783) compared to the Female group (0.944946 and 0.932722, respectively), indicating that male predictions have slightly better discrimination and reliability. Nevertheless, females have a higher recall (0.535558) and FLAG (0.680424), which implies that he/she can better capture true churners with a slight increase in the false positives. The small Brier score among males (0.074351) is also lower than that of females (0.095941), which is another indication of superior probability calibration in the male subgroup.

Stability and distributional consistency

Figure 8 overlays the predicted churn probabilities for the training set (blue) and test set (orange). The diagram provides a quick check for drift: if the model is applied to data with a significantly different score distribution, thresholds, profit curves, and calibration may become invalid. In Fig. 8, the distributions are similar; both peaks are sharp at 0.12–0.20 (most customers are low risk) with a long right tail to 0.8. This stability enables the use of a single global threshold and consistent cost curve assumptions, eliminating the need for immediate retuning if future overlays diverge—especially with mass shifting to higher or lower risk—re-run calibration and profit analysis to update the budget and threshold.

Decision and lift curve analysis

Figure 9 illustrates an experimental view of the proposed model, where campaign revenue is adjusted as the decision threshold for flagging customers. Expected profit becomes strongly negative on the left side of the curve when almost everyone gets flagged. The rise in the threshold sharply increases the profit, then flattens at the decision threshold of 0.436. During this period, fewer high-risk customers were contacted under the assumed economics of \$4,414. Beyond that point, the benefit drops faster than the costs, leading to a decline in profit and identifying many true churners due to the strict threshold. Figure 9 suggests zero profit at a threshold of 0.8. Positive profit remains with 0.268 precision and 0.661 recall. The diagram suggests maximising profit by adjusting the threshold through re-estimation of the cost curve for budget efficiency.

Decision curve Analysis in Fig. 10 maintains a dashed green line, representing the zero baseline, where ‘Treat None’ means no one is contacted, and the net profit is zero for all thresholds. ‘Treat all’ declines in the orange

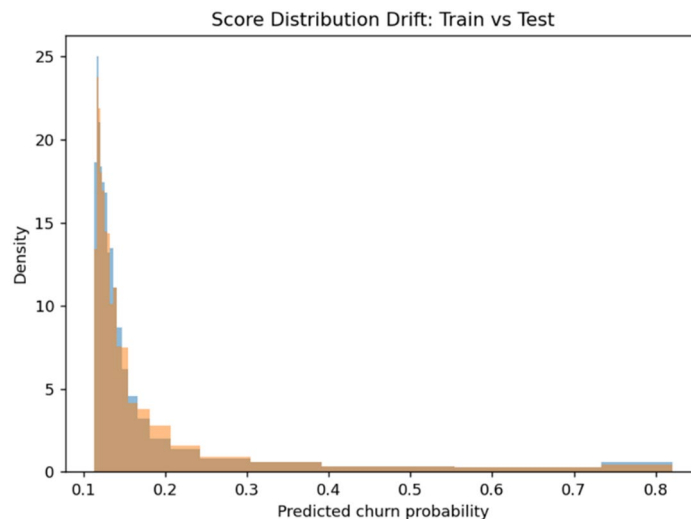


Fig. 8. Distributional consistency of churn risk scores across training and evaluation datasets.

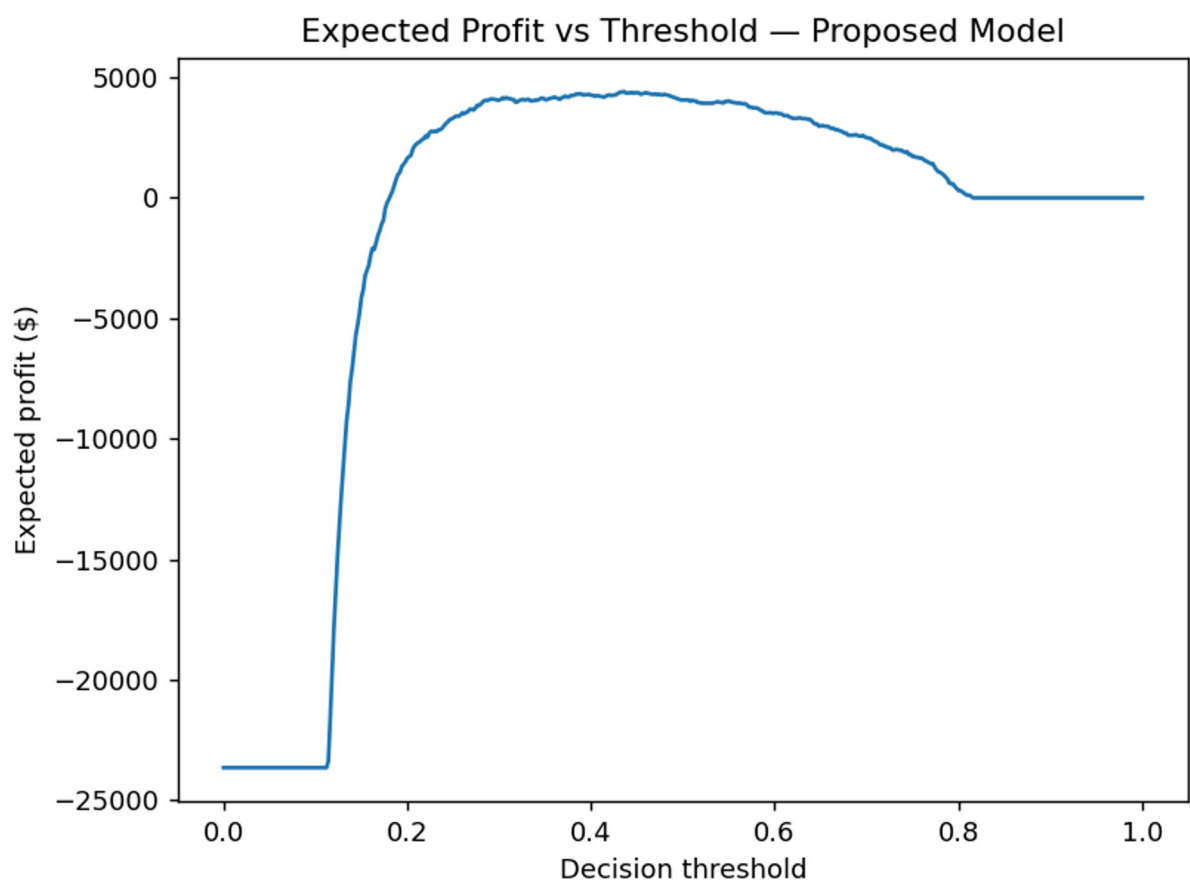


Fig. 9. Profit-maximising threshold curve for the proposed model.

dashed line as the threshold increases, suggesting that blindly contacting everyone is harmful. The blue lines represent the net benefit of using the proposed model to determine whom to contact, showing the advantage of this approach in correctly identifying true churners, compared to treating everyone or no one, after accounting for false positive penalties.

$$\text{Expected Benefit} = \text{value} - \text{per} - \text{retained} \times \text{retention} - \text{effect} \quad (12)$$

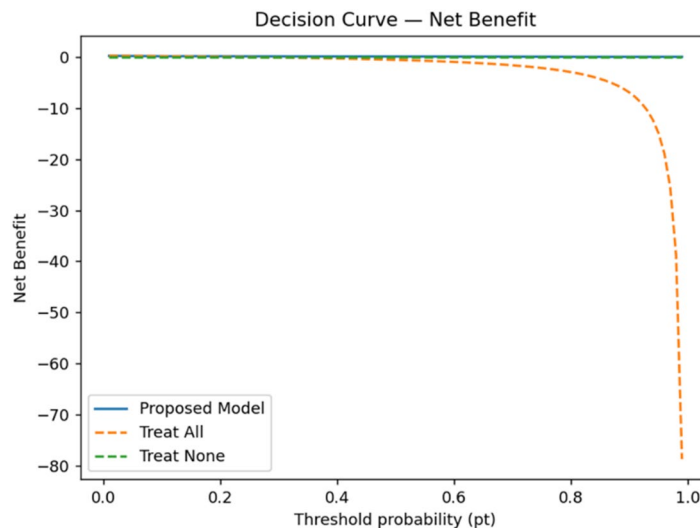


Fig. 10. Decision curve showing threshold-dependent net benefit versus baseline strategies.

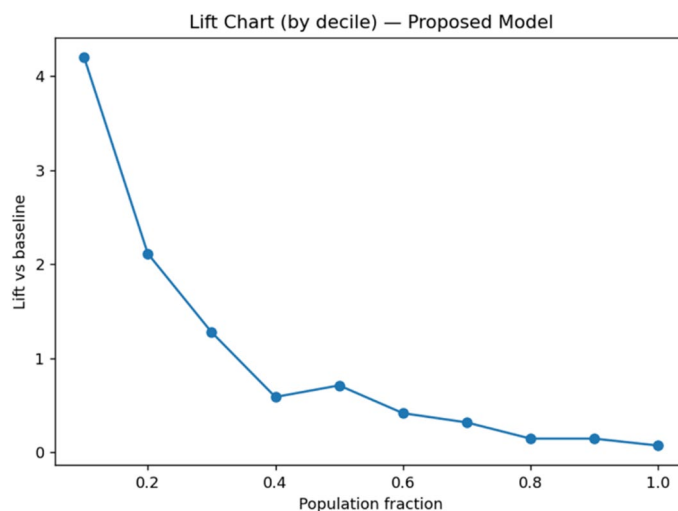


Fig. 11. Decile-wise lift curve showing concentrated churn detection in top-ranked segments.

In equation, value – per – retained is a scenario-level estimation taken as 200 during this estimation. The contacted churning is encoded as a probability using retention – effect. Direct incentive or operational cost formulated through cumulative cost,

$$\text{Cumulative Cost } (k) = k \times (\text{offer}_{\text{cost}} + \text{contact}_{\text{cost}}) \quad (13)$$

Finally, the cumulative profit curve evaluated via,

$$\text{Cumulative}_{\text{profit}} = \text{Cumulative}_{\text{benefit}} - \text{Cumulative Cost } (k) \quad (14)$$

This Lift Chart (by decile) highlights how well the Proposed Model concentrates true churners at the top of the ranked customer list. In Fig. 11, the x-axis represents the cumulative share of the targeted top 10%-20% population, and the y-axis is lift—the churn rate within that slice divided by the overall churn rate. The curve starts around 4.2X lift in the first decile, meaning the top 10% flagged by the model contains over four times as many churners as a random 10% sample—excellent targeting power.

Limitations

Even with the significant improvements presented by the discretized encoding and standard scaling, there are a number of drawbacks to the churn prediction system, which is based on the neural network. To begin with, it is common with models to have issues with compliance with data privacy, bias in the developed algorithm, and

explainability of AIs, which make such solutions hazardous and unacceptable concerning ethical and regulatory standards in business settings⁴². Second, the use of historical data sets limits the dynamically responsive character of the model towards quickly changing customer behavior and transforming market environments, as well as reduces its long-term flexibility and resiliency⁴³. Thirdly, most of the studies carried out have focused on statistical performance measures, including accuracy and F1-score, without taking into consideration business constraints and deployment practicability. This results in a disconnect between research results and practical application, and is frequently poorly operationalised to provide models assumed to be theoretically sound, but models guiding practical implementation that are hard to weave into any existing workflows. Additionally, the deep neural network architectures, such as RNNs and DNNs, exhibit high accuracy, but they tend to be uninterpretable and require more computational units, and this is why the simplified models, such as the Random Forests, are an appropriate choice for production applications since they are easily trained and understood. These observations help to amplify the reason why a compromise should be taken towards the adoption of a balanced approach, one that is able to compromise about the predictive performance with one that can analyze and take into account other factors, such as interpretability, calculational efficiency, and other integration-related factors. Future research, therefore, should support the creation of adaptive predictive churn models, which can adapt to the latest types of churns, accept even superior-quality feature engineering, and embed pre-trained predictors to improve generalisation and performance. Also, by examining the variants of deep learning, such as convolutional and recurrent neural networks, it could become possible to extract the population patterns of complex behavior more effectively based on various types of sources of data using the variants, which will further increase the accuracy of predictions.

Future research directions

Further research would be required to focus on the combination of real-time data stream together with dynamic model systems to predict churn patterns, thus churn prediction systems would be more agile and adaptable to the nature of a rapidly changing environment⁴³. In addition to the demonstration of the representatives of the previous category, the emergence of deep learning-based designs, namely those that are adjusted to work with imbalanced and unstructured data, may enhance the pace at which the dynamics of churn patterns and subtle customer retention regulations may be detected. Further research on hybrid modelling techniques to hybridise traditional machine learning functions with expert engineers can allow the presentation of domain information that will improve inferential and practical functionality of model inventions. Parallel Ensemble learning models and boosters may augment prediction capability and robustness, particularly where the data that are available on the coefficients of class imbalance is high. In addition, there must be advanced data, transformation, and feature selection techniques to ascertain the consistency of the best model functionality as well as its practicality in capturing the most effective variables under usage in the churn of the client. Research on the condition of metaheuristic optimization models (via genetic algorithms, particle swarm optimization, and ant colony optimization) can also contribute to assisting hyperparameters more accurately, and thus improve the generalization of the model in different groups of customers. In the future, imbalance-aware parameters with DART boosters must be included to extend the search space. The confidence interval must be recorded via Friedman test statistics. The strategies of the feature engineering alongside the employment of the cost-sensitive-learning frameworks provide a glimpse of hope in exposing the hidden tendencies of behavior and internal associations in the complex data⁴⁴. The feature interpretability and reducing noise and redundancy can be enhanced with the help of such techniques as impact encoding high-cardinality categorical variables and strong attribute selection. Finally, evolutionary-based/swarm-based, bio-inspired dimensionality reduction algorithms can also efficiently be implemented so as to reduce the computational load but increase the scalability of high-dimensional, high-scale data repositories.

Conclusion

The study establishes that incorporating a powerful categorical encoding scheme and efficient scaling of a neural network architecture can boost customer churn prediction performance in a great way, especially when dealing with information of very high dimensions and heterogeneity. The model tested on a complex dataset had a high precision (0.8528) and a competitive accuracy (0.8660) and ROC-AUC (0.8657), which is valuable since this restricts the number of false positives, which is important in customer retention. This was comparable and, in some aspects, better than time-tested methods of machine learning like Gradient Boosting and the Random Forest, which, although with slightly better F1-scores, could not match the precision and generalization of the neural network in more complicated settings.

Moreover, cross-validation and group heterogeneity, which offer stability and the flexibility of the model on the platform of demographic and geographic ranking, are an alternative to such models that provide powerful generalization as well as convenient conduct of such models in the development of Customer Relations Management measures. The significance value of the features has displayed that two factors have been presented as the most significant predictors in line with the balance of churn quality and status of active member, where finance engagement and involvement have proven to be core predictors of customer retention.

Data availability

The authors declare that the data supporting the findings of this study are available within the manuscript.

Received: 5 November 2025; Accepted: 21 January 2026

Published online: 27 January 2026

References

- Liu, Y. & Zhuang, Y. Research model of churn prediction based on customer segmentation and misclassification cost in the context of big data. *J. Comput. Communication*. 3(6), 87–93. <https://doi.org/10.4236/jcc.2015.36009> (2015).
- Jahromi, A. T., Sepehri, M. M., Teimourpour, B. & Choobdar, S. Modeling customer churn in a non-contractual setting: the case of telecommunications service providers. *J. Strategic Mark.* 18(7), 587–598. <https://doi.org/10.1080/0965254x.2010.529158> (2010).
- Dhotre, S., Galange, S., Awasarmol, S., Patil, M. & Malule, R. Customer retention analysis for Telecom industry. *Int. J. Res. Appl. Sci. Eng. Technol.* 11(11), 1484–1488. <https://doi.org/10.22214/ijraset.2023.56804> (2023).
- Banu, J. F. et al. Artificial intelligence-based customer churn prediction model for business markets. *Comput. Intell. Neurosci.* <https://doi.org/10.1155/2022/1703696> (2022).
- Vafeiadis, T., Diamantaras, K., Sarigiannidis, G. & Chatzivasavvas, K. A comparison of machine learning techniques for customer churn prediction. *Simul. Model. Pract. Theory*. 55, 1–9. <https://doi.org/10.1016/j.simpat.2015.03.003> (2015).
- Ebrah, K. & Elnasir, S. Churn prediction using machine learning and recommendation plans for telecoms. *J. Comput. Commun.* 7(11), 33–53. <https://doi.org/10.4236/jcc.2019.711003> (2019).
- Breiman, L. Random forests. *Mach. Learn.* 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324> (2001).
- Friedman, J. H. Greedy function approximation: A gradient boosting machine. *Ann. Stat.* 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451> (2001).
- Chen, T. & Guestrin, C. XGBoost: A scalable tree boosting system. in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining* 785–794 (2016). <https://doi.org/10.1145/2939672.2939785>
- Sikri, A., Jameel, R., Idrees, S. M. & Kaur, H. Enhancing customer retention in Telecom industry with machine learning driven churn prediction. *Sci. Rep.* 14(1). <https://doi.org/10.1038/s41598-024-63750-0> (2024).
- Khattak, A. et al. Customer churn prediction using composite deep learning technique. *Sci. Rep.* 13(1), 17294. <https://doi.org/10.1038/s41598-023-44396-w> (2023).
- Alghamdi, N. A. & Al-Baity, H. H. Augmented analytics driven by AI: A digital transformation beyond business intelligence. *Sensors* 22(20), 8071. <https://doi.org/10.3390/s22208071> (2022).
- Ong, J., Tong, G., Khor, K. & Haw, S. Enhancing customer churn prediction with resampling: A comparative study. *TEM J.* 1927–1936. <https://doi.org/10.18421/tem133-20> (2024).
- Peng, K., Peng, Y. & Li, W. Research on customer churn prediction and model interpretability analysis. *PLoS ONE*. 18(12), e0289724. <https://doi.org/10.1371/journal.pone.0289724> (2023).
- Chajia, M. & Nfaoui, E. H. Customer churn prediction approach based on LLM embeddings and logistic regression. *Future Internet*. 16(12), 453. <https://doi.org/10.3390/fi16120453> (2024).
- Ngai, E., Xiu, L. & Chau, D. Application of data mining techniques in customer relationship management: A literature review and classification. *Expert Syst. Appl.* 36(2), 2592–2602. <https://doi.org/10.1016/j.eswa.2008.02.021> (2009).
- Goodfellow, I., Bengio, Y. & Courville, A. *Deep Learning* (MIT Press, 2016). <https://doi.org/10.5555/3086952>
- Liang, Z. Efficient representations for high-cardinality categorical variables in machine learning. *ArXiv* <https://doi.org/10.48550/arxiv.2501.05646> (2025).
- Mirkovic, M., Lolic, T., Stefanovic, D., Anderla, A. & Gracanin, D. Customer churn prediction in B2B non-contractual business settings using invoice data. *Appl. Sci.* 12(10), 5001. <https://doi.org/10.3390/app12105001> (2022).
- Sahar, F. N. Machine-learning techniques for customer retention: A comparative study. *Int. J. Adv. Comput. Sci. Appl.* 9(2). <https://doi.org/10.14569/ijacsa.2018.090238> (2018).
- Malikireddy, V. P. & Kasa, M. Customer churn prediction model based on machine learning techniques: A systematic review. in *Proc. 3rd Int. Conf. Integrated Intelligent Computing Communication & Security (ICIIC)* 167–174 (2021). <https://doi.org/10.2991/aic.k.210913.021>
- Bhuria, R. et al. Ensemble-based customer churn prediction in banking: A voting classifier approach for improved client retention using demographic and behavioral data. *Discover Sustain.* 6(1). <https://doi.org/10.1007/s43621-025-00807-8> (2025).
- Srinivasan, R., Rajeswari, D. & Elangovan, G. Customer churn prediction using machine learning approaches. in *Proc. Int. Conf. Artif. Intell. Knowl. Discovery Concurrent Engineering (ICECONF)* 1–6 (2023). <https://doi.org/10.1109/ICECONF57129.2023.100838139>
- Chinnaraj, R. Bio-inspired approach to extend customer churn prediction for the Telecom industry in efficient way. *Wirel. Pers. Commun.* 133(1), 15–29. <https://doi.org/10.1007/s11277-023-10697-6> (2023).
- Matteucci, F., Arzamasov, V. & Boehm, K. A benchmark of categorical encoders for binary classification. *ArXiv* <https://doi.org/10.48550/arxiv.2307.09191> (2023).
- Imani, M., Joudaki, M., Beikmohammadi, A. & Arabnia, H. R. Customer churn prediction: A review of recent advances, trends, and challenges in conventional machine learning and deep learning. *Preprints* <https://doi.org/10.20944/preprints202503.1969.v1> (2025).
- Chang, V. et al. Prediction of customer churn behavior in the telecommunication industry using machine learning models. *Algorithms* 17(6), 231. <https://doi.org/10.3390/a17060231> (2024).
- Li, N. A., Yang, T., Zhan, X., Shi, Y. & Li, H. Utilizing data science and AI for customer churn prediction in marketing. *J. Theory Pract. Eng. Sci.* 4(5), 72–79. [https://doi.org/10.53469/jtpes.2024.04\(05\).10](https://doi.org/10.53469/jtpes.2024.04(05).10) (2024).
- Maan, J. & Maan, H. Customer churn prediction model using explainable machine learning. *ArXiv* <https://doi.org/10.48550/arxiv.2303.00960> (2023).
- Alotaibi, M. Z. & Haq, M. A. Customer churn prediction for telecommunication companies using machine learning and ensemble methods. *Eng. Technol. Appl. Sci. Res.* 14(3), 14572–14578. <https://doi.org/10.48084/etasr.7480> (2024).
- Md, A., Hossain & Islam, M. S. A novel hybrid feature selection and ensemble-based machine learning approach for botnet detection. *Sci. Rep.* 13(1). <https://doi.org/10.1038/s41598-023-48230-1> (2023).
- Huynh, Q. T., Nguyen, T. T. & Nguyen, H. Base resistance of super-large and long piles in soft soil: performance of artificial neural network model and field implications. *Acta Geotech.* 18(5), 2755. <https://doi.org/10.1007/s11440-022-01736-w> (2022).
- Liu, X., Xia, G., Zhang, X., Ma, W. & Yu, C. Customer churn prediction model based on hybrid neural networks. *Sci. Rep.* <https://doi.org/10.1038/s41598-024-79603-9> (2024).
- Srikanth, M. V., Sunitha, P., Puppala, R. S., Asileti, S. K. & Akshaykranth, A. A novel framework for intrusion detection in IOT networks using hybrid optimization algorithm and convolutional neural networks. *Frankl. Open.* 14, 100461. <https://doi.org/10.1016/j.fraope.2025.100461> (2026).
- Sai Poojitha, C. et al. *A Learning Model on Customer Churn Forecasting for Telecom Providers*. IC3T 2024. *Lecture Notes in Networks and Systems* Vol. 1357 (Springer, 2026). https://doi.org/10.1007/978-981-96-7477-0_2
- Santos, V. C. A., Cardoso, L. F. F. & Alves, R. The quest for the reliability of machine learning models in binary classification on tabular data. *Sci. Rep.* 13(1). <https://doi.org/10.1038/s41598-023-45876-9> (2023).
- Vaiyapuri, T. et al. Metaheuristics based dimensionality reduction with deep learning driven false data injection attack detection for enhanced network security. *Sci. Rep.* 14(1). <https://doi.org/10.1038/s41598-024-69806-5> (2024).
- Yan, Y. & Resnick, N. A high-performance turnkey system for customer lifetime value prediction in retail brands. *Quant. Market. Econ.* 22(2), 169. <https://doi.org/10.1007/s11129-023-09272-x> (2023).
- Basal, N. M., Moula, N. K. & Cetin, N. A. Predictive analytics for customer behavior prediction in artificial intelligence. *Econ. World.* 12(2). <https://doi.org/10.17265/2328-7144/2025.02.006> (2025).
- Verma, B. D., Pratap, R. & Bera, D. Efficient binary embedding of categorical data using binsketch. *Data Min. Knowl. Discov.* 36(2), 537–565. <https://doi.org/10.1007/s10618-021-00815-y> (2022).

41. De, S., Prabu, P. & Paulose, J. Application of machine learning in customer churn prediction, in *Innovations Power Adv. Comput. Technol. (i-PACT)* 1–7 (2021). <https://doi.org/10.1109/i-pact52855.2021.9696771>
42. Gaidhani, Y. et al. AI-driven predictive analytics for CRM to enhance retention personalization and decision-making. *Int. J. Adv. Comput. Sci. Appl.* **16**(4). <https://doi.org/10.14569/ijacsa.2025.0160456> (2025).
43. Googerdchi, K. F., Asadi, S. & Jafari, S. M. Customer churn modeling in telecommunication using a novel multi-objective evolutionary clustering-based ensemble learning. *PLoS ONE*. **19**(6), e0303881. <https://doi.org/10.1371/journal.pone.0303881> (2024).
44. Manteigas, C. & António, N. Understanding and predicting lapses in mortgage life insurance using a machine learning approach. *Expert Syst. Appl.* **255**, 124753. <https://doi.org/10.1016/j.eswa.2024.124753> (2024).

Acknowledgements

The authors would like to express their gratitude to the Symbiosis International University for the funding arrangements.

Author contributions

Brijit Bhattacharjee, Utpal Madhu, Sutirtha Kumar Guha, Somasree Bhadra, Pritam Kumar Das contributed to conceptualization and methodology. Subham Pankaj Samantaray, M. Zubairuddin and Shahid Tamboli contributed to writing & proof reading.

Funding

Open access funding provided by Symbiosis International (Deemed University).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.T.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026